

SAPHUL PMS

Live Flow Explanation

1. System Start

- Backend runs FastAPI at `http://localhost:8000`
- Frontend runs React/Vite at `http://localhost:5173`
- On load, frontend checks for saved JWT token in local storage
- Unauthenticated users are redirected to the login page

2. Login Flow

- User enters email and password
- Frontend sends credentials to backend
- Backend validates and returns JWT token + permission list
- Frontend stores token, user details, and permissions
- User is redirected to the dashboard

3. Signup Flow

- New user opens signup page and enters details
- Backend creates the account
- User is automatically logged in and redirected to the app

4. Forgot & Reset Password

- User enters registered email on forgot password page
- Backend generates a reset token
- User opens reset link, enters and confirms new password
- Backend validates token and updates the password

5. Dashboard

- User lands on dashboard after login
- Backend returns statistics scoped to user's access level
- Admin sees all projects and tasks system-wide
- PM, Member, and Client see only their assigned data

6. Project Creation (Admin)

- Admin opens Projects page and clicks Add Project
- Fills in name, description, methodology, dates
- Selects Project Manager and initial members
- Backend creates project with PENDING status
- Activity log records the creation

7. Task Creation (Admin / PM)

- PM or Admin opens project and clicks Add Task
- Selects assignee, enters title, description, dates, priority, estimated hours
- Backend verifies project access and creates the task
- Task appears in task list with TODO status
- Activity log records the creation

8. Member Task Update

- Member opens their assigned task
- Member updates status (TODO → IN_PROGRESS → DONE)
- Backend confirms the user is the assigned member
- Progress auto-updates: TODO=0%, IN_PROGRESS=50%, DONE=100%
- Member cannot edit title or description
- Backward status movement blocked once task is done or under review

9. Task Approval Flow

- Member sets task to DONE and clicks Request Approval
- Task moves to Under Review with Approval Pending status
- PM or Admin reviews the task
- Approve: progress set to 100%, status marked Approved
- Reject: review and approval status marked Rejected
- All actions are logged in the activity log

10. Project Completion Flow

- PM requests project completion
- Project moves to COMPLETED with pending review

- Admin reviews and approves or rejects
- On approval: project status becomes CLOSED
- On rejection: review status becomes Rejected

11. Client Invitation Flow

- Admin or PM enters client email from project details page
- Backend generates secure invite token (valid 7 days)
- Client receives invitation email with link
- Client opens link, sets password
- Backend creates client account if needed, assigns Client role
- Client is added to the project and can now log in
- Client sees only their assigned project — no global access

12. File Upload Flow

- User opens project details and selects a file to upload
- Backend saves file in local uploads folder and records metadata
- If marked as latest, previous latest files are unmarked
- File appears in project file list
- Activity log records the upload

13. Activity Log

- Every major action is auto-logged (create, update, delete, approve, reject, upload)
- Viewable from project details page
- Shows who performed the action and when

14. End-to-End Demo Sequence

- 1. Login as Admin → view dashboard statistics
- 2. Create a project → assign PM and members
- 3. Open project → view members, tasks, files, activity
- 4. Create a task → assign to a member
- 5. Login as Member → update task to DONE → request approval
- 6. Login as PM → approve or reject the task
- 7. Upload a project file → view in file list
- 8. Invite a client → client accepts → logs in with restricted view
- 9. Request project completion → login as Admin → approve

15. Key Business Value

- Centralized project and task tracking in one system
- Clear role separation: Admin, Project Manager, Member, Client
- Permission-controlled access to prevent accidental data exposure
- Approval workflows for both task completion and project closure
- Full activity history for accountability
- File versioning for project and task documents
- Soft delete to preserve all business records